MSc Project - Reflective Essay

PROJECT TITLE	**GAIT ANALYSIS USING MACHINE LEARNING CLASSIFIERS**
STUDENT NAME	**JUMMA MOHAMMAD TELI**
STUDENT NUMBER	**231034661**
SUPERVISOR NAME	**TAYYAB AHMAD ANSARI**
PROGRAMME OF STUDY	**MSc INTERNET OF THINGS AND FUTURE NETWORKS**

Introduction

My research detail involved applying several machine learning classifiers on gait patterns through marker-based motion capture data. The main objective of this study was to identify the different walking speeds, which are a very essential indication of a person's health. This study applied several machine learning models, including Random Forest, K-Nearest Neighbors, and Support Vector Machines and Neural Networks, among others, in establishing the most efficient approach for the classification of gait speed. Gait analysis has been imperative, in both clinical and research settings, for the diagnosis of movement disorders, rehabilitation progress monitoring, and therapeutic intervention evaluation; however, the size of the developed data is huge and complex in nature. Therefore, this study has tried to overcome these challenges by using advanced machine learning techniques.

1. The Difference Between Theory and Practical Work

In theory, models like Random Forest, SVM, and Neural Networks have proved efficient in pattern recognition and classification tasks, particularly ones that include highly dimensional data. Results for these models have always turned out very impressive in areas like image and speech recognition. However, when these very models were applied in gait analysis, several practical issues emerged: the high dimensionality and noise in the motion capture data highly affected the complications of training these models, hence affecting their accuracy.

For instance, while the Random Forest model was able to theoretically achieve an accuracy of 0.99, because of controlled environments, it performed differently in real-life conditions where the data was either noisy or incomplete. Although the SVM is very powerful in handling high-dimensional data, it could not manage the size of the dataset and, as such, needed some chunking methods for training and evaluation. Though practical, this added another layer of complication, mostly in terms of ensuring that the learning would be the same throughout the dataset. These challenges underline very well the difference between theoretical performance and real-world application—imperfections in real-world data play a large role in model performance.

2. Strengths and Weaknesses of the Approach

One of the major strengths of the study was that it comprehensively assessed several machine learning classifiers. In a scientific comparison, it provided insight into what each of these classifiers—Random Forest, k-NN, SVM, and Neural Networks—could do to classify gait speed. For example, the model developed from Random Forest always ensured high precision, recall, and F1-scores across all categories of speed, hence being an accurate model. The k-NN classifier also performed very well, especially in discriminating fine differences in walking speed.

The study also revealed some weaknesses. On the other hand, the Naive Bayes classifier was overwhelmed by the high dimensionality of the data and hence performed very badly with immense misclassifications. Although the Logistic Regression models, even with L1 and L2 regularization, could not correctly classify the middle speed classes, thereby exposing their weakness toward the complexity of gait data. A Neural Network, though powerful, was computationally very intensive and prone to overfitting with noisy data. Moreover, the chunking technique applied to SVM, even while being a requirement for large datasets, was prone to inconsistencies in the learning process because of training data division into chunks that might impact the model's overall performance.

3. Challenges Encountered and Solutions Implemented

One of the most influential complications in conducting this research was related to high dimensionality and noise in the motion capture data. The complexity of the data contributed to either overfitting or poor performance of the models in practical scenarios. In this respect, feature engineering played an important role since it provided key features that would help the models better capture the essence of gait mechanics, hence improving their performance.

Another challenge related to training the models on large datasets was very computationally demanding, specifically with regard to neural networks and SVM models. This is where the chunking method implemented for the SVM training comes into play—it will enable the model to process data in portions without overwhelmed computational resources. But it also added complexity in terms of ensuring every chunk's representativeness of overall dataset characteristics, which will be further handled carefully with data splitting and validation strategies.

4. Learning and Development

In the project, I had major insights into how machine learning can be practically applied in a complex, real-world environment. One of the essential items learned was how to improve model performance through data preprocessing and feature engineering. Theoretically, one learns about machine learning algorithms; however this project built on me that input data quality mostly controls the outcome.

This project has also enhanced my problem-solving skills in terms of adjusting theoretical models to practical challenges. Those involved the practical application of the chunking method for SVM and managing a balance between model accuracy and computational efficiency—good learning experiences. All of these challenges taught me once again that flexibility and innovation are principles to be followed when working with machine learning techniques on real-world problems.

5. Future Work

If more time is given in the future, several areas would be addressed to improve the performance of the models and their applicability. First of all, handling data imbalance and noise is needed. The use of advanced feature engineering or even deep learning models with data augmentation techniques would enhance the robustness and accuracy in the models. Besides, an important future development in a clinical setting would be the integration of real-time gait analysis to provide immediate feedback and adjustment in the process of rehabilitation.

Another future research direction would be toward the exploration of ensemble methods and hybrid models, where different classifiers are combined. For instance, the robustness of

random forests could be married to the precision of SVM or the adaptability of neural networks to find better performance across different gait speed categories. Further ahead, a dataset created from a greater variety of subjects and walking conditions would provide a more rounded assessment of the generalizability of the models.

6. Ethical Considerations in Gait Analysis Using AI

Several ethical considerations should be taken into consideration with respect to AI in the gait analysis, especially on issues related to data privacy and bias, but most importantly in its potential impact on patient care. First, it is important to ensure that any motion capture data used in model training are appropriately anonymized and stored securely so that the privacy of the patients is guaranteed. Algorithmic bias can be an actual big problem if not managed. Unbiased outputs of the models are most dependent on representative data used to train, unless biased outputs mean misdiagnosis or unequal treatment within a clinical setting.

While AI models can offer valuable insights, they really should not replace human judgment in any way, especially in clinical decision-making. Models are just tools for assistance but never stand-alone decision-makers. Wanting to have transparency into how models work and giving them the possibility of human oversight is very key to observing ethical standards within health.

7. Broader Impact and Future Potential

Such research has a broader impact on gait analysis, including other areas of human movement analysis, for example, sports science, biomechanics, and rehabilitation. Methodologies and models built in this context will then be able to adjust to analyze different kinds of movement patterns, provide insights into various health conditions and health-related issues, and increase the efficacy of any rehabilitation program. Future studies could also work on the potentials for their integration into other emerging technologies, such as augmented reality and virtual reality, for creating immersive and interactive rehabilitation settings that would provide advanced and engaging environments for patient care.

8. Personal Development

This project has been a source of immense growth for me, both personally and professionally. Dealing with complex machine learning algorithms and their application in the gait analysis domain has increased my technical depth. Besides, my critical thinking and problem-solving skills were developed very much. The iterative nature of the project helped me realize the importance of patience and adaptability in research by continuously enhancing models and techniques.

I also developed a stronger appreciation for the importance of interdisciplinary knowledge. The project required me to close the gap between the disciplines of computer science and biomechanics; it involved a deep understanding of both so that machine learning techniques would be relevantly applied to gait analysis. This gave me a broader perspective and more holistic view of how technology can be applied to really complex healthcare problems.

My communication skills have improved, especially on how to communicate complex technical issues to a non-technical audience. Report writing, presentation of findings, and discussing the challenges faced with my supervisor and peers enhanced my capability to express complex ideas in a clear and concise manner. This skill will be of great benefit in my future career, where effective communication across disciplines and with different levels of technical stakeholders is indispensable.

Another area of personal development was time management and project planning. Planning and managing the demands of data preprocessing, model training, and result analysis called for a very strict use of time. This experience has enabled me to do better in managing large-scale projects and ensuring each phase is efficiently done while maintaining a high standard of work.

Conclusion

In conclusion, this research work has been a great opportunity to learn and contributes much toward my growth, both as a person and academically. In the study, it was pointed out that machine learning classifiers could be used in classifying walking speed. However, it was found that Random Forest, k-Nearest Neighbours, and Support Vector Machines are the best classifiers. The project mainly showed concrete difficulties in the study of theoretical models over real data: noise, size, and lack of computing resources.

The results of this study apply not only to the field of gait analysis but also allow a view into a more general perspective, human movement analysis. Moreover, successful completion of this project underlines feature engineering, model evaluation, and an effective use of machine learning in healthcare. Thirdly, the ethical considerations that emanated from the study appear to underline needs and best practices for making sure that technologies unfolding either way cannot compromise quality patient care and equity.

What excites me most about this looking forward is being in a position to go ahead and continue doing exploration at that junction of technology and health, which this project has built on in terms of skill and knowledge acquisition. Machine learning offers tremendous potential for changing health, so I feel good about making a contribution within this growing, dynamic, and impactful field. This has fitted quite a good educational base, and I am quite positive that the lessons learned are likely to guide my future endeavors in both academic research and professional practice.

References

- Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.
- Mannini, A., & Sabatini, A. M. (2010). Machine learning methods for classifying human physical activity from on-body accelerometers. Sensors, 10(2), 1154-1175.
- Munoz-Ospina, P., et al. (2022). Impact of Artificial Intelligence on Gait Analysis. Journal of Rehabilitation Research and Development, 59(3), 445-456.
- Witten, I. H., Frank, E., & Hall, M. A. (2011). Data Mining: Practical Machine Learning Tools and Techniques. Elsevier.
- Mangasarian, O. L., & Thompson, M. E. (2008). Chunking for massive nonlinear kernel classification. *Optimization Methods and Software*, *23*(3), 365–374. https://doi.org/10.1080/10556780701611976.
- Witten, I. H., Frank, E., & Hall, M. A. (2011). *Data Mining: Practical Machine Learning Tools and Techniques* (3rd ed.). Morgan Kaufmann.